PRESTO-R: Predictive, Stability-Aware Reinforcement Learning Routing

Abstract

This report evaluates risk-aware routing with a GRU link-degradation predictor and a switch-penalized DQN policy. Results are from the repository's simulated network environment; they are not evidence of deployment performance on production networks. All reported metrics and tables below are generated from saved test episodes and resolved configurations.

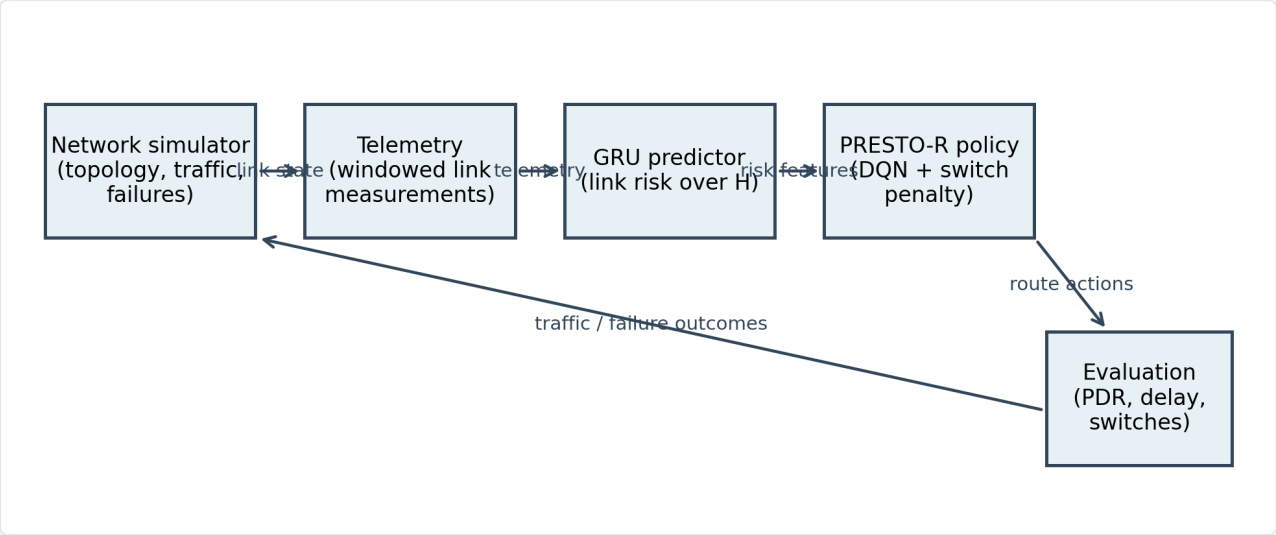
1. Introduction

Routing policies must balance delivery and recovery against the instability caused by frequent path changes. PRESTO-R augments per-path observations with predicted link risk and adds a penalty for switching. The project evaluates whether this combination improves routing behavior over a reactive DQN, a predictive policy without the selected switch penalty, and a shortest-path failover baseline.

2. Related work and scope

The experimental comparisons are internal baselines implemented in this repository. The project does not claim a reproduction of an external RQSMR implementation. Its predictor is trained on simulator-derived telemetry, and failures are synthetic. Claims in this report are limited to the tested simulator and seed ranges.

3. System design



PRESTO-R architecture

The simulator supplies noisy telemetry and route outcomes. A GRU estimates link degradation risk over horizon H; the DQN policy observes path features and risk features. Its reward includes the selected switch penalty λ. Heuristic and learned policies use a common deterministic rollout and metric pipeline.

4. Experimental setup

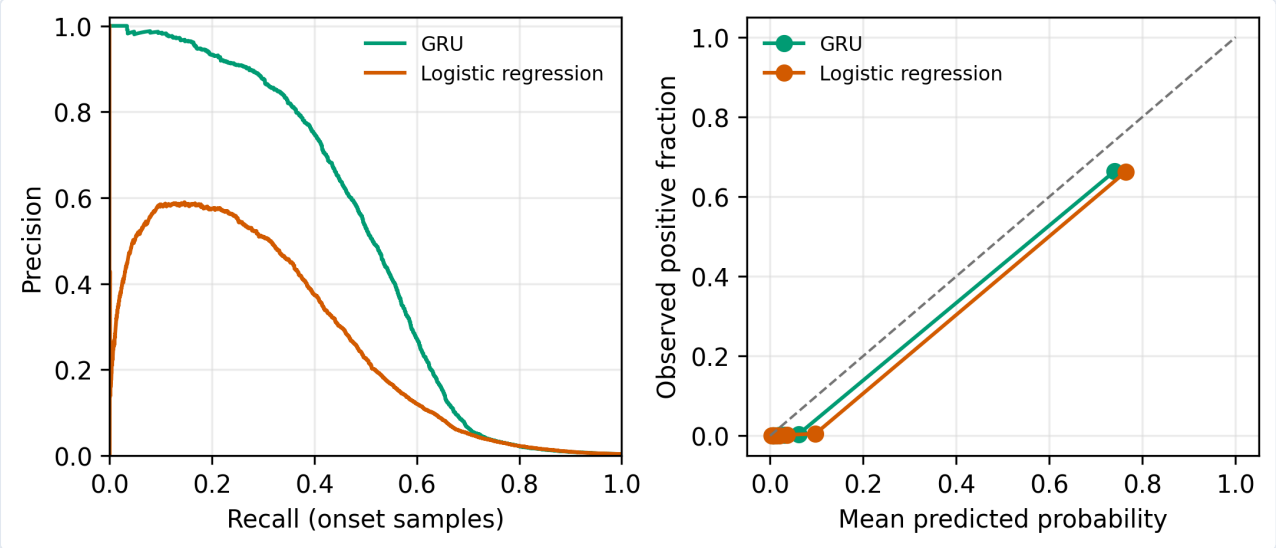
The main held-out grid uses five scenarios and two topologies. NSFNET learned agents use five training seeds; RGG agents use three. The common test-seed range and configuration-derived parameters are shown below. PDR is delivered traffic divided by offered demand; recovery time, proactive rate, and route switches are computed per episode. Across-seed standard deviations describe training-seed variation; paired tests use matched held-out scenario seeds with Holm correction.

T1. Simulation and training parameters

Parameter	Value
Primary topology	NSFNET
Episode length (ticks)	200
Link capacity / base-delay range (ms)	10.0 / [2, 12]
Candidate paths	4
Predictor window W / horizon H	10 / 5
Switch penalty λ	0.5
Agent training steps	300000
Training seeds	0–4 (NSFNET); 0–2 (RGG)
Held-out test seeds	10000–10199
Scenarios	steady, congestion, node_failure, fluctuation, mixed

5. Results

5.1 Predictor quality

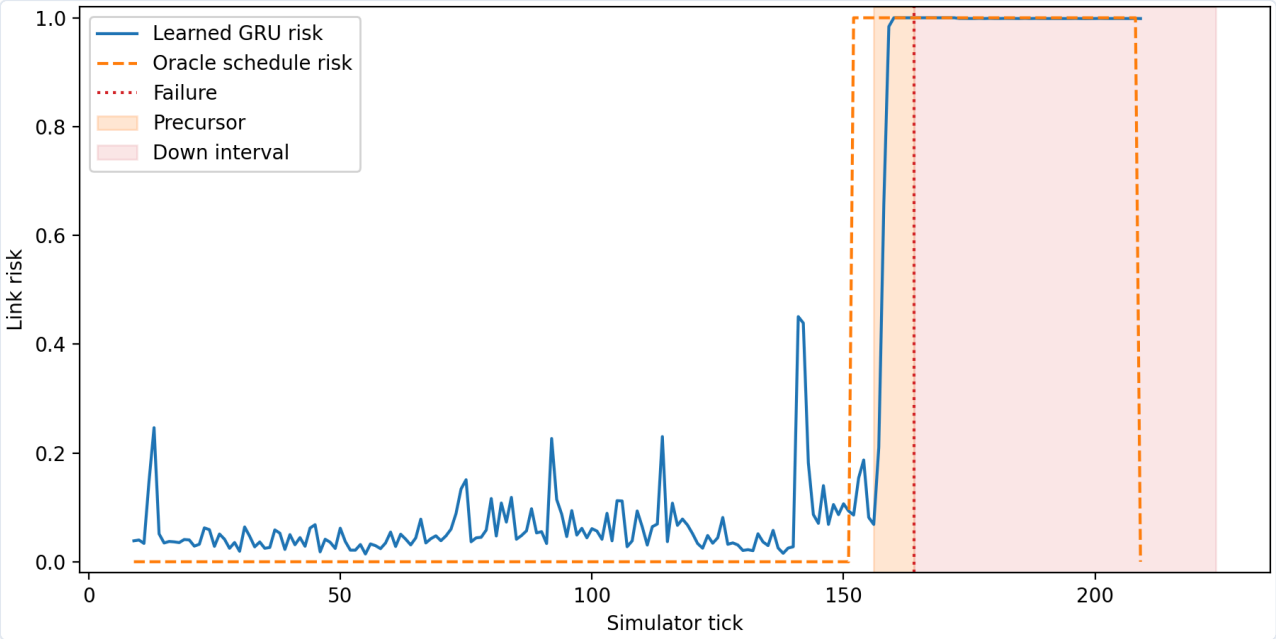


Predictor precision-recall and calibration

The precision-recall panel is restricted to onset samples. Calibration uses the held-out predictor test split. The table distinguishes onset from all samples and includes predictor subgroup rows when available.

T2. Held-out predictor metrics

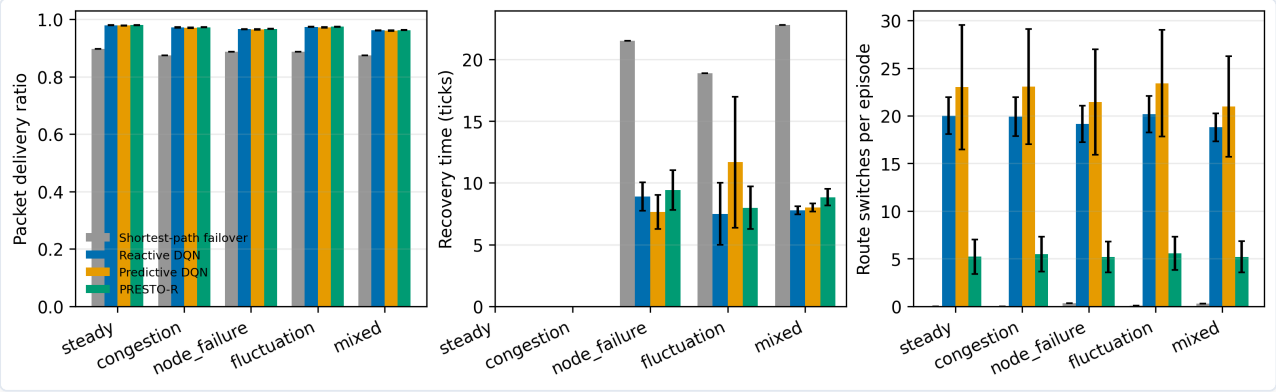
model	subset	group	f1	pr_auc	brier	median_lead_time	samples
gru	all	all	0.9769	0.9861	0.0054	3.0	822705
gru	onset	all	0.5296	0.4945	0.0054	3.0	769718
gru	onset	failure:precursor	0.7476	0.8639	0.3343	—	1256
gru	onset	failure:sudden	0.6667	0.5388	0.4615	—	240
logreg	all	all	0.9699	0.9836	0.0075	1.0	822705
logreg	onset	all	0.3949	0.2656	0.0075	2.0	769718
logreg	onset	failure:precursor	0.7534	0.8022	0.3566	—	1256
logreg	onset	failure:sudden	0.661	0.5348	0.4516	—	240



Example learned and oracle risk trace

The risk trace is a deterministic illustrative mixed-scenario episode, not an aggregate score.

5.2 Main comparison



NSFNET results

On NSFNET mixed test episodes, PRESTO-R averaged 5.20 switches per episode versus 20.99 for Predictive DQN (75.2% fewer) and its PDR was 0.9625 versus 0.9608 (difference +0.0017). These values average episode metrics within each training seed before averaging across training seeds.

Bars show the mean across training-seed episode averages; error bars are standard deviations across training seeds. The deterministic heuristic has no between-training-seed variation.

T3. NSFNET test results (mean ± standard deviation over training seeds)

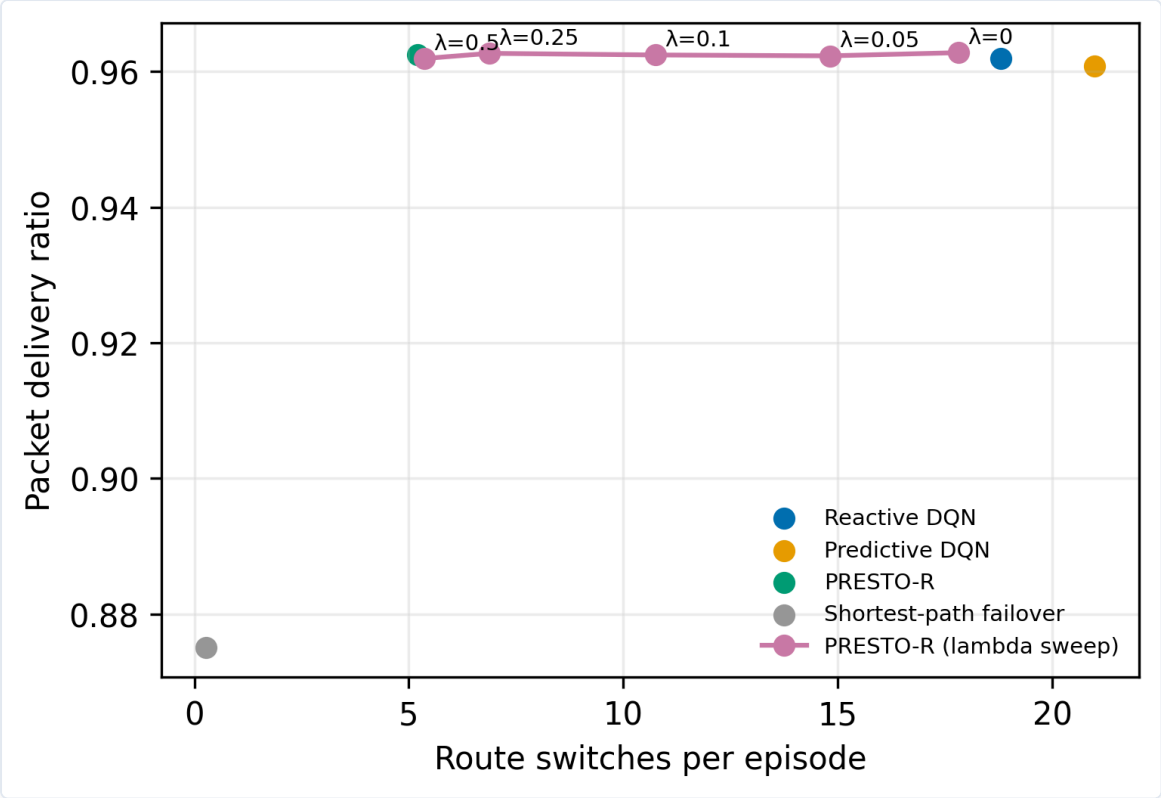
Method	Scenario	pdr	delay_mean	recovery_time	switches	proactive_rate	unnecessary_switch_rate
sp_failover	steady	0.8966 ± NA	119.9542 ± NA	not available	0.0000 ± NA	not available	not available
sp_failover	congestion	0.8747 ± NA	128.1184 ± NA	not available	0.0000 ± NA	not available	not available
sp_failover	node_failure	0.8871 ± NA	117.6611 ± NA	21.5272 ± NA	0.3100 ± NA	0.0068 ± NA	0.0000 ± NA
sp_failover	fluctuation	0.8875 ± NA	124.2878 ± NA	18.8750 ± NA	0.0700 ± NA	0.0000 ± NA	0.0000 ± NA
sp_failover	mixed	0.8751 ± NA	126.5853 ± NA	22.8085 ± NA	0.2650 ± NA	0.0000 ± NA	0.0000 ± NA
reactive_dqn	steady	0.9797 ± 0.0009	46.3062 ± 0.7003	not available	20.0070 ± 1.9388	not available	0.4127 ± 0.0349
reactive_dqn	congestion	0.9721 ± 0.0009	52.6087 ± 0.6362	not available	19.9130 ± 2.0395	not available	0.3818 ± 0.0225
reactive_dqn	node_failure	0.9662 ± 0.0010	52.6682 ± 0.8414	8.8951 ± 1.1546	19.1490 ± 1.9134	0.5070 ± 0.0331	0.3675 ± 0.0284
reactive_dqn	fluctuation	0.9740 ± 0.0011	49.7406 ± 0.6025	7.4906 ± 2.5128	20.1760 ± 1.9200	0.2886 ± 0.0470	0.3907 ± 0.0245
reactive_dqn	mixed	0.9619 ± 0.0008	56.6613 ± 0.7133	7.7741 ± 0.3258	18.7990 ± 1.4575	0.4572 ± 0.0339	0.3444 ± 0.0279
predictive_dqn	steady	0.9786 ± 0.0008	46.7965 ± 1.5823	not available	23.0240 ± 6.5355	not available	0.4975 ± 0.0390
predictive_dqn	congestion	0.9708 ± 0.0010	53.0682 ± 1.4765	not available	23.0640 ± 6.0548	not available	0.4524 ± 0.0247
predictive_dqn	node_failure	0.9652 ± 0.0013	53.4841 ± 1.3807	7.6455 ± 1.3806	21.4570 ± 5.5372	0.5872 ± 0.0444	0.4429 ± 0.0200
predictive_dqn	fluctuation	0.9726 ± 0.0011	50.7261 ± 1.6293	11.6746 ± 5.3045	23.4190 ± 5.6024	0.2294 ± 0.0928	0.4656 ± 0.0215
predictive_dqn	mixed	0.9608 ± 0.0013	57.1976 ± 0.9776	8.0037 ± 0.3307	20.9880 ± 5.2802	0.5108 ± 0.0578	0.4128 ± 0.0342
presto_dqn	steady	0.9804 ± 0.0007	46.2766 ± 1.0454	not available	5.2120 ± 1.8193	not available	0.2836 ± 0.0585
presto_dqn	congestion	0.9729 ± 0.0006	52.4105 ± 0.8563	not available	5.4780 ± 1.8468	not available	0.2794 ± 0.0614
presto_dqn	node_failure	0.9667 ± 0.0007	52.5669 ± 0.8969	9.4236 ± 1.5982	5.1820 ± 1.6214	0.4472 ± 0.0426	0.2412 ± 0.0500

Method	Scenario	pdr	delay_mean	recovery_time	switches	proactive_rate	unnecessary_switch_rate
presto_dqn	fluctuation	0.9749 ± 0.0007	49.8912 ± 1.1183	7.9852 ± 1.7305	5.5570 ± 1.7518	0.2418 ± 0.0564	0.2766 ± 0.0575
presto_dqn	mixed	0.9625 ± 0.0005	56.3287 ± 0.9610	8.8419 ± 0.6810	5.1970 ± 1.6276	0.3901 ± 0.0524	0.2344 ± 0.0494

T4. Paired PRESTO-R comparisons on NSFNET

scenario	metric	baseline	n_episodes	median_difference	p_holm	effect_size_r
congestion	pdr	predictive_dqn	200	3.989e-06	8.57e-10	0.482
congestion	pdr	reactive_dqn	200	0	0.914	0.000
congestion	switches	predictive_dqn	200	-7.8	8.38e-22	-0.713
congestion	switches	reactive_dqn	200	-4.7	1.63e-20	-0.691
fluctuation	pdr	predictive_dqn	200	2.4945e-06	1.47e-06	0.399
fluctuation	pdr	reactive_dqn	200	0	0.0104	0.000
fluctuation	recovery_time	predictive_dqn	16	0	1	0.000
fluctuation	recovery_time	reactive_dqn	18	0	1	0.000
fluctuation	switches	predictive_dqn	200	-8.9	8.86e-23	-0.728
fluctuation	switches	reactive_dqn	200	-4.1	3.05e-21	-0.704
mixed	pdr	predictive_dqn	200	2.302e-05	4.77e-05	0.354
mixed	pdr	reactive_dqn	200	0	0.616	0.000
mixed	recovery_time	predictive_dqn	58	0	1	0.000
mixed	recovery_time	reactive_dqn	59	0	0.472	0.000
mixed	switches	predictive_dqn	200	-8.1	1.2e-25	-0.772
mixed	switches	reactive_dqn	200	-6.3	1.15e-23	-0.742
node_failure	pdr	predictive_dqn	200	5.4865e-06	0.00154	0.301
node_failure	pdr	reactive_dqn	200	0	1	0.000
node_failure	recovery_time	predictive_dqn	53	0	0.0493	0.000
node_failure	recovery_time	reactive_dqn	51	0	0.262	0.000
node_failure	switches	predictive_dqn	200	-8	2.02e-23	-0.739
node_failure	switches	reactive_dqn	200	-6.2	3.41e-22	-0.719
steady	pdr	predictive_dqn	200	0	3.52e-08	0.000
steady	pdr	reactive_dqn	200	0	0.341	0.000
steady	switches	predictive_dqn	200	-4.1	9.87e-19	-0.661
steady	switches	reactive_dqn	200	-2.2	2.21e-17	-0.637

5.3 Prediction and switch-penalty ablation



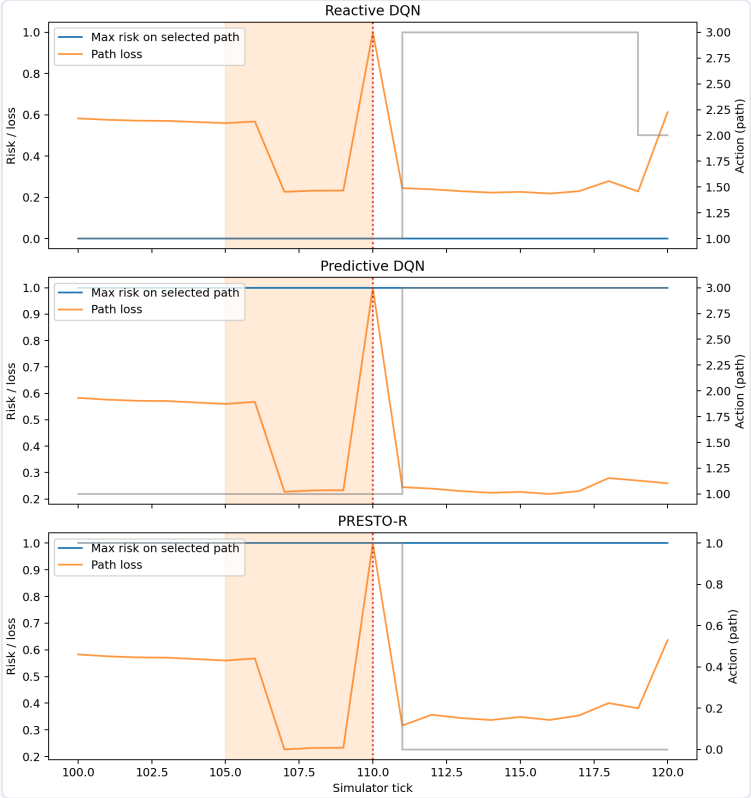
Switch penalty trade-off

The λ curve uses mixed-scenario test episodes for the validation-screened penalty variants. The full 2×2 factorial is not available because the no-prediction/penalty (`reactive_dqn_pen`) cell was not part of the completed P0 test grid; it is explicitly marked rather than inferred.

T5. Available mixed-scenario ablation cells

Ablation	Method	PDR (mean over seeds)	Switches / episode (mean over seeds)
No prediction; no penalty	reactive_dqn	0.9619	18.799
Prediction; no penalty	presto_lam0	0.9628	17.812
Prediction; selected penalty	presto_dqn	0.9625	5.197
No prediction; selected penalty	reactive_dqn_pen	not run in the completed P0 grid	not run in the completed P0 grid

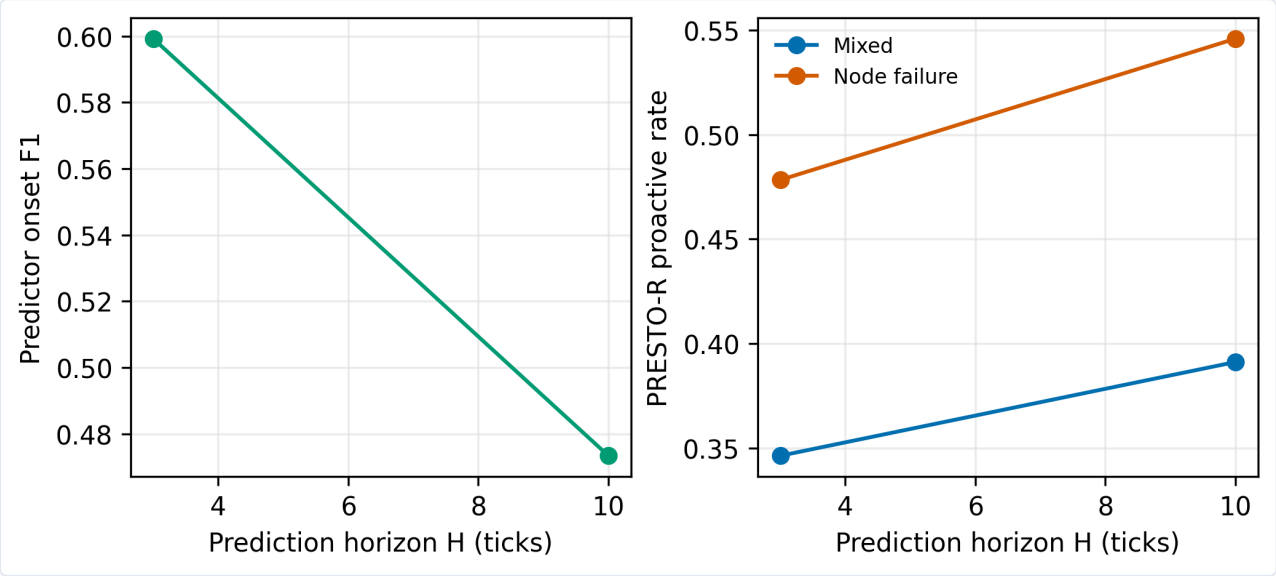
5.4 Behavior around failures



Held-out mixed-scenario event trace

The event trace shows the first suitable held-out test episode found among the first 20 test seeds. It is illustrative and should be interpreted alongside the per-episode statistics, not as a representative estimate.

5.5 Sensitivity to horizon and switch penalty



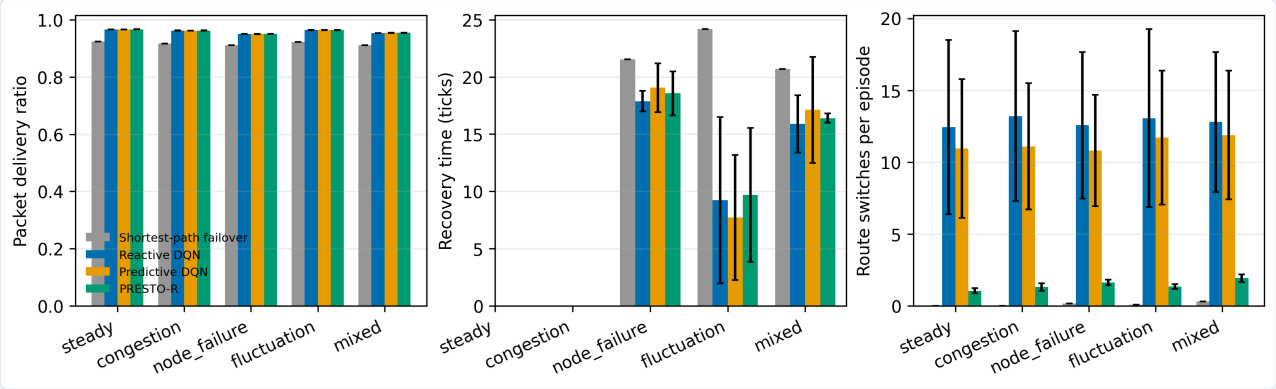
Prediction horizon sensitivity

Horizon-specific agents are evaluated on mixed and node-failure test scenarios; predictor F1 is measured on the corresponding held-out predictor data. The λ sweep is shown in F5.

Horizon-sweep results

horizon	scenario	pdr_mean	pdr_std	switches_mean	switches_std	proactive_rate_mean	proactive_rate_std	recove
3	mixed	0.9613	0.0016	6.8733	2.0496	0.3463	0.0653	9.7458
3	node_failure	0.9661	0.0017	6.675	1.9609	0.4783	0.0352	9.3308
10	mixed	0.9616	0.0015	5.5583	1.8525	0.3912	0.0171	8.8698
10	node_failure	0.9658	0.0019	5.8983	2.2085	0.5459	0.0096	9.2278

5.6 Second topology and robustness



RGG results

RGG test results

Method	Scenario	pdr	delay_mean	recovery_time	switches	proactive_rate	unnecessary_switch_rate
sp_failover	steady	0.9242 ± NA	68.3566 ± NA	not available	0.0000 ± NA	not available	not available
sp_failover	congestion	0.9170 ± NA	70.6413 ± NA	not available	0.0000 ± NA	not available	not available
sp_failover	node_failure	0.9116 ± NA	68.4587 ± NA	21.5526 ± NA	0.1550 ± NA	0.0000 ± NA	0.0000 ± NA
sp_failover	fluctuation	0.9230 ± NA	68.1102 ± NA	24.1667 ± NA	0.0650 ± NA	0.0000 ± NA	0.0000 ± NA
sp_failover	mixed	0.9108 ± NA	71.3116 ± NA	20.6959 ± NA	0.2850 ± NA	0.0058 ± NA	0.0000 ± NA
reactive_dqn	steady	0.9665 ± 0.0003	34.6919 ± 0.3298	not available	12.4383 ± 6.0721	not available	0.6090 ± 0.1239
reactive_dqn	congestion	0.9625 ± 0.0003	35.6699 ± 0.3392	not available	13.1983 ± 5.9101	not available	0.6268 ± 0.0945
reactive_dqn	node_failure	0.9505 ± 0.0003	36.8101 ± 0.2686	17.8846 ± 0.9031	12.5583 ± 5.1001	0.3853 ± 0.0151	0.6007 ± 0.0860
reactive_dqn	fluctuation	0.9643 ± 0.0004	35.6269 ± 0.4518	9.2333 ± 7.2590	13.0617 ± 6.1823	0.0667 ± 0.0577	0.5861 ± 0.0971
reactive_dqn	mixed	0.9542 ± 0.0002	37.9024 ± 0.3205	15.8897 ± 2.5097	12.7883 ± 4.8685	0.3438 ± 0.0343	0.5656 ± 0.0596
predictive_dqn	steady	0.9664 ± 0.0004	35.2924 ± 0.4202	not available	10.9417 ± 4.8419	not available	0.6496 ± 0.1253
predictive_dqn	congestion	0.9625 ± 0.0002	36.3546 ± 0.6216	not available	11.0900 ± 4.4003	not available	0.6616 ± 0.0919
predictive_dqn	node_failure	0.9504 ± 0.0003	37.2256 ± 0.2825	19.0515 ± 2.1342	10.8100 ± 3.8714	0.3817 ± 0.0398	0.6129 ± 0.1033
predictive_dqn	fluctuation	0.9647 ± 0.0005	36.0663 ± 0.3574	7.7180 ± 5.4765	11.7017 ± 4.6742	0.2968 ± 0.0847	0.6378 ± 0.0972
predictive_dqn	mixed	0.9546 ± 0.0007	38.4670 ± 0.3982	17.1224 ± 4.6365	11.8733 ± 4.4776	0.4826 ± 0.1154	0.6037 ± 0.0747
presto_dqn	steady	0.9669 ± 0.0003	34.4730 ± 1.4464	not available	1.0383 ± 0.1922	not available	0.3146 ± 0.0184
presto_dqn	congestion	0.9627 ± 0.0005	35.5245 ± 1.5505	not available	1.3033 ± 0.2642	not available	0.3733 ± 0.0195
presto_dqn	node_failure	0.9508 ± 0.0004	36.5316 ± 1.0023	18.5616 ± 1.9305	1.6250 ± 0.1922	0.3607 ± 0.0478	0.2928 ± 0.0337

Method	Scenario	pdr	delay_mean	recovery_time	switches	proactive_rate	unnecessary_switch_rate
presto_dqn	fluctuation	0.9647 ± 0.0005	35.5024 ± 1.6147	9.6881 ± 5.8649	1.3267 ± 0.1692	0.0859 ± 0.0834	0.3274 ± 0.0148
presto_dqn	mixed	0.9545 ± 0.0010	38.1368 ± 1.9417	16.3996 ± 0.3937	1.9133 ± 0.2644	0.4159 ± 0.0706	0.2932 ± 0.0256

T6. Evaluation-only robustness checks

case	method	metric	mean_over_training_seeds	std_over_training_seeds	training_seeds
sudden	reactive_sudden	pdr	0.9619	0.0008	5
sudden	reactive_sudden	switches	18.799	1.4575	5
sudden	reactive_sudden	recovery_time	7.7741	0.3258	5
sudden	reactive_sudden	proactive_rate	0.4572	0.0339	5
sudden	presto_sudden	pdr	0.9625	0.0005	5
sudden	presto_sudden	switches	5.197	1.6276	5
sudden	presto_sudden	recovery_time	8.8419	0.681	5
sudden	presto_sudden	proactive_rate	0.3901	0.0524	5
noise2x	reactive_noise2x	pdr	0.9618	0.0007	5
noise2x	reactive_noise2x	switches	19.916	1.3194	5
noise2x	reactive_noise2x	recovery_time	8.0025	0.471	5
noise2x	reactive_noise2x	proactive_rate	0.4418	0.0494	5
noise2x	presto_noise2x	pdr	0.9626	0.0005	5
noise2x	presto_noise2x	switches	5.582	1.6968	5
noise2x	presto_noise2x	recovery_time	9.0004	0.8428	5
noise2x	presto_noise2x	proactive_rate	0.3994	0.0599	5
load150	reactive_load150	pdr	0.8728	0.001	5
load150	reactive_load150	switches	18.846	2.1709	5
load150	reactive_load150	recovery_time	16.1781	0.2633	5
load150	reactive_load150	proactive_rate	0.4341	0.0262	5
load150	presto_load150	pdr	0.8736	0.0013	5
load150	presto_load150	switches	4.985	1.8631	5
load150	presto_load150	recovery_time	17.038	1.9265	5
load150	presto_load150	proactive_rate	0.3972	0.0388	5

Sudden failures remove precursor warning; noise and background-load changes are applied only at evaluation. No robustness condition is used for retraining or test-based tuning.

Mixed-scenario λ sensitivity data

lambda	method	pdr_mean	pdr_std	switches_mean	switches_std	unnecessary_switch_rate_mean	unnecessary_sw
0.0	presto_lam0	0.9628	0.0697	17.8125	25.7984	0.376	0.3446
0.05	presto_lam005	0.9623	0.071	14.8125	24.591	0.3562	0.3407
0.1	presto_lam01	0.9624	0.0705	10.7475	17.7226	0.3167	0.3147
0.25	presto_lam025	0.9627	0.0703	6.8725	12.9127	0.2486	0.288
0.5	presto_lam05	0.9619	0.0725	5.3625	9.5133	0.2585	0.2963

6. Discussion and limitations

Results describe a fluid simulator rather than packet-level network behavior. The setup controls one managed flow and uses synthetic failures with designed precursor behavior. The predictor is trained on telemetry generated under the simulator's failover policy, which can limit transfer to other policies or networks. The sudden-failure check is especially important because such events cannot be anticipated from precursor signals. The no-prediction/penalty factorial ablation is missing, and the report does not treat the available partial ablation as a complete causal decomposition.

7. Conclusion and future work

PRESTO-R combines risk features with a switching cost; the outcomes and failed criteria are reported without suppressing negative comparisons. Further validation should add multiple managed flows, multipath splitting, packet-level or real-network traces, and online predictor updates. NS-3 validation remains optional Phase 7 work and is not included here.

8. Success criteria

T7. Predeclared criteria

Check	Result	Details
PRESTO-R vs reactive in node_failure	FAIL	PDR 0.9667 vs 0.9662; recovery 9.440 vs 8.915; Holm p PDR=1, recovery=0.2617
PRESTO-R vs reactive in fluctuation	FAIL	PDR 0.9749 vs 0.9740; recovery 7.844 vs 7.386; Holm p PDR=0.01042, recovery=1
PRESTO-R vs predictive in mixed	PASS	switches 5.197 vs 20.988; PDR 0.9625 vs 0.9608
PRESTO-R vs predictive in fluctuation	PASS	switches 5.557 vs 23.419; PDR 0.9749 vs 0.9726
Steady-state delay	PASS	PRESTO-R delay 46.2766; limit 50.9369 (reactive × 1.10)
Predictor onset F1	FAIL	GRU validation onset F1=0.5296; threshold=0.7000
Reproducibility	PASS	Step and event parquet hashes match on a fresh rerun

Reproducibility

From the repository root, install the pinned dependencies, then run `make quick` for a smoke validation. With the completed raw Phase 5 results present, `make tables`, `make figures`, and `make report` regenerate the derived outputs. All full-grid runs can be resumed with the Phase 5 commands in the README. Test outputs must not be used to retune models or thresholds.